

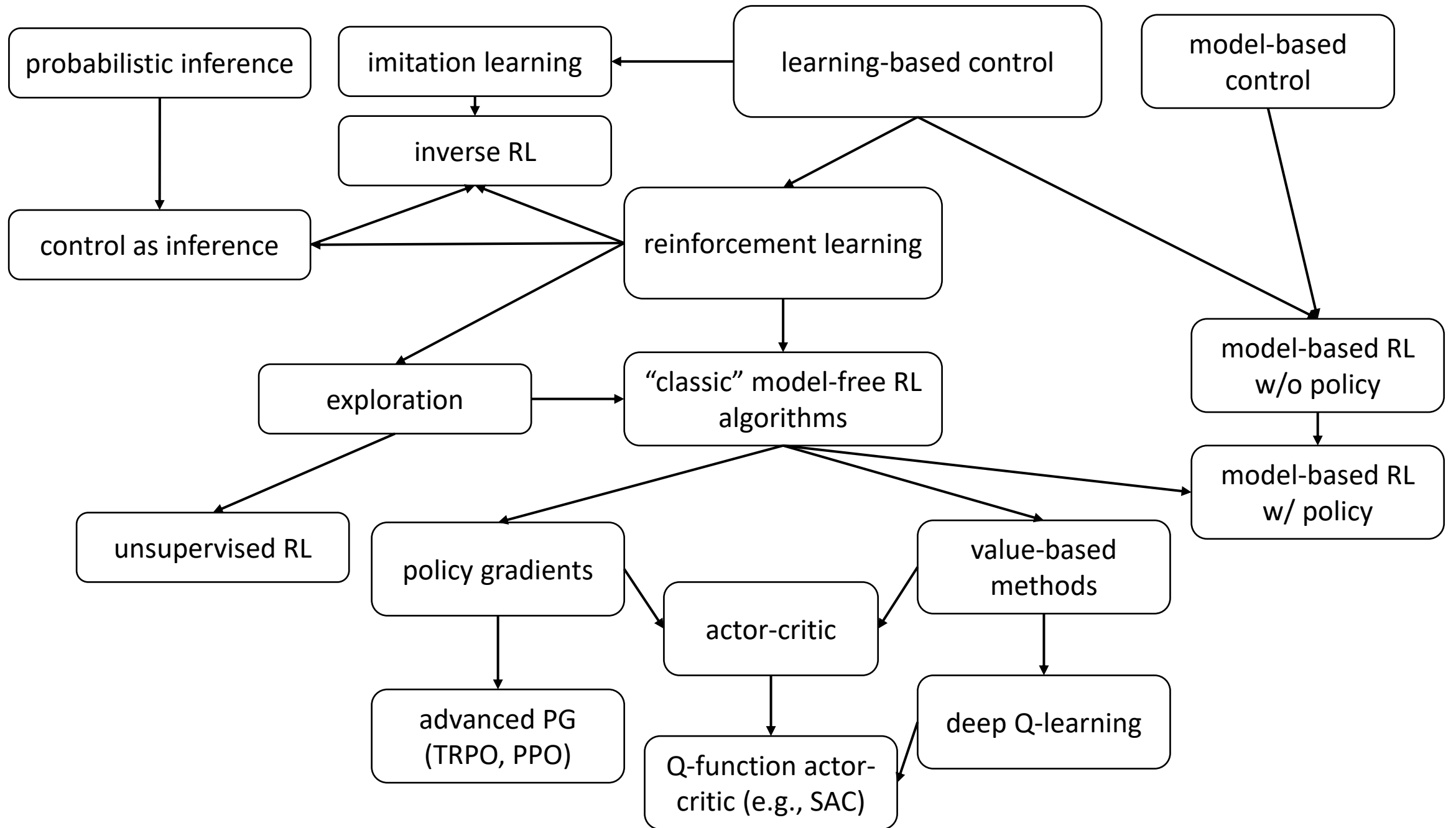
Challenges and Open Problems

CS 185/285

Instructor: Sergey Levine
UC Berkeley



A Brief Review



Why do we have so many RL algorithms?

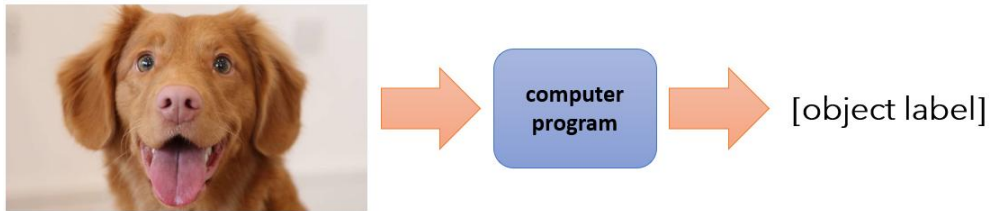
What is reinforcement learning?

Mathematical formalism for learning-based
decision making

Approach for learning decision making and control
from experience

What is reinforcement learning?

supervised learning



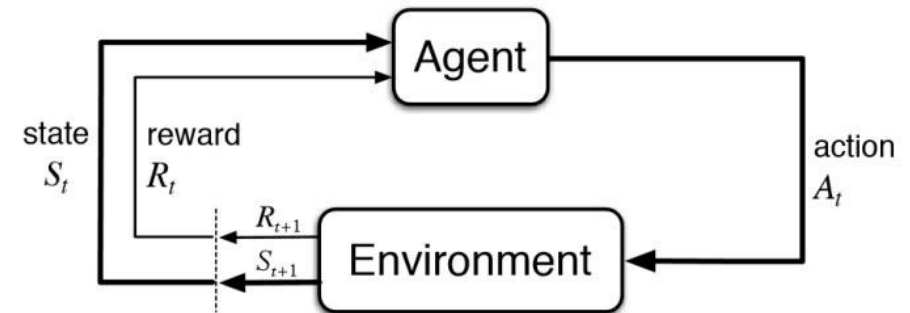
input: $\mathbf{x}$

output: $\mathbf{y}$

data: $\mathcal{D} = \{(\mathbf{x}_i, \mathbf{y}_i)\}$

goal: $f_{\theta}(\mathbf{x}_i) \approx \mathbf{y}_i$

reinforcement learning



input: $\mathbf{s}_t$ at each time step

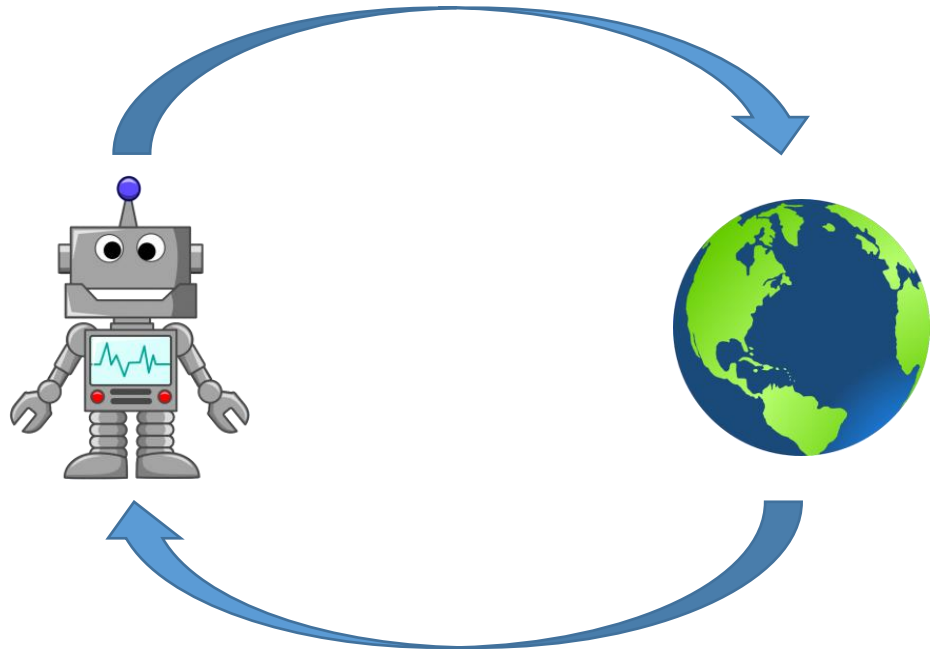
output: $\mathbf{a}_t$ at each time step

data: $(\mathbf{s}_1, \mathbf{a}_1, r_1, \dots, \mathbf{s}_T, \mathbf{a}_T, r_T)$

goal: learn $\pi_{\theta} : \mathbf{s}_t \rightarrow \mathbf{a}_t$

to maximize $\sum_t r_t$

decisions (actions)



consequences
observations (states)
rewards



Actions: muscle contractions
Observations: sight, smell
Rewards: food



Actions: motor current or torque
Observations: camera images
Rewards: task success measure (e.g., running speed)



Actions: what to purchase
Observations: inventory levels
Rewards: profit

Some examples...

Advanced Skills by Learning Locomotion and Local Navigation End-to-End

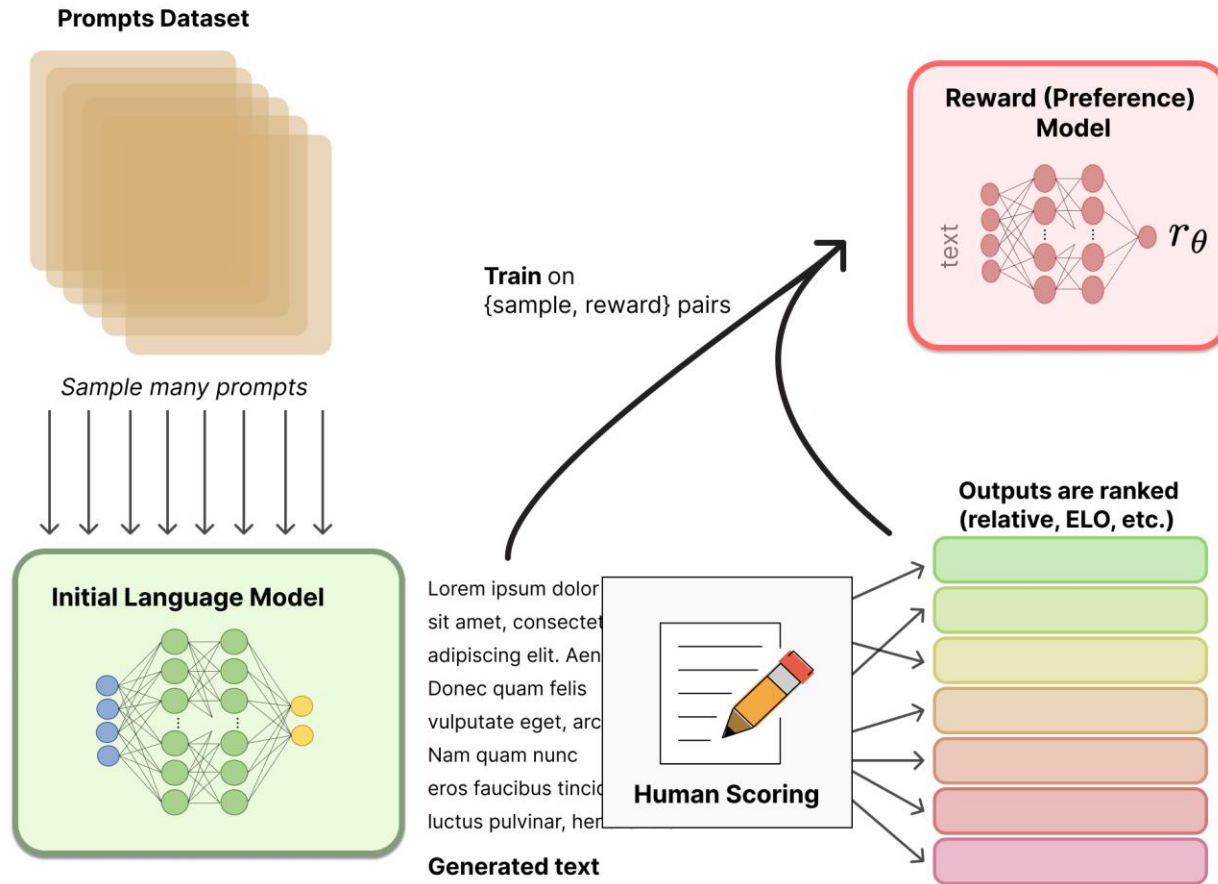
Nikita Rudin, David Hoeller,
Marko Bjelonic and Marco Hutter



Some examples...



Some examples...



LE

Can you explain to me how RL with human feedback works for language models like I am five?



Sure! So, imagine you have a robot named Rufus who wants to learn how to talk like a human. Rufus has a language model that helps him understand words and sentences.

First, Rufus will say something using his language model. For example, he might say "I am a robot."

Then, a human will listen to what Rufus said and give him feedback on whether it sounded like a natural sentence a human would say. The human might say, "That's not quite right, Rufus. Humans don't usually say 'I am a robot.' They might say 'I'm a robot' or 'I am a machine.'"

Rufus will take this feedback and use it to update his language model. He will try to say the sentence again, using the new information he received from the human. This time, he might say "I'm a robot."

The human will listen again and give Rufus more feedback. This process will continue until Rufus can say sentences that sound natural to a human.

Over time, Rufus will learn how to talk like a human thanks to the feedback he receives from humans. This is how language models can be improved using RL with human feedback.

Some perspectives...

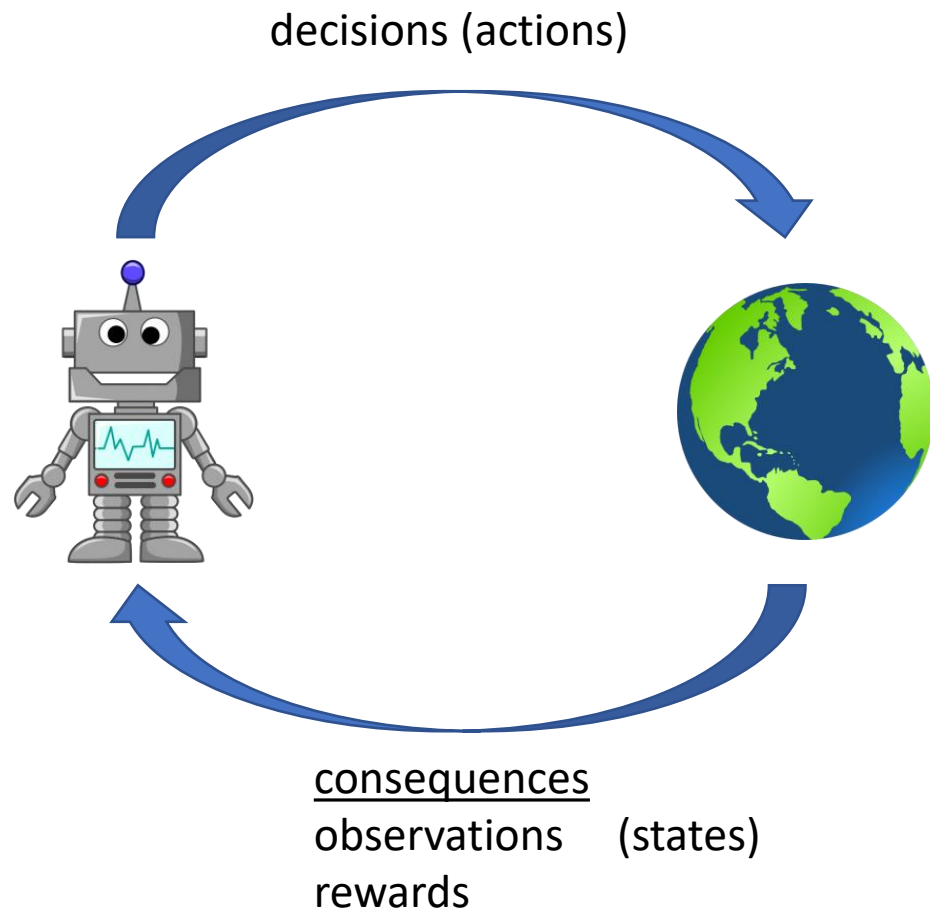
Reinforcement Learning as an Engineering Tool

Reinforcement Learning and the Real World

Reinforcement Learning as “Universal” Learning

Reinforcement Learning as an Engineering Tool

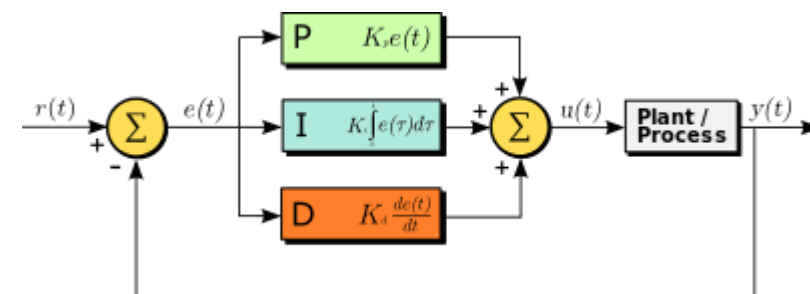
What we think RL is...



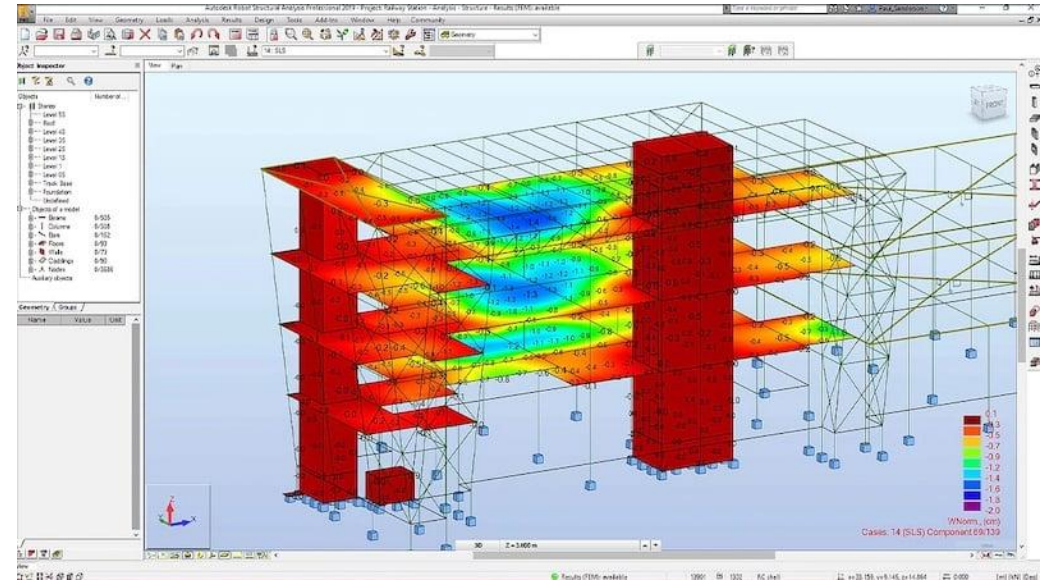
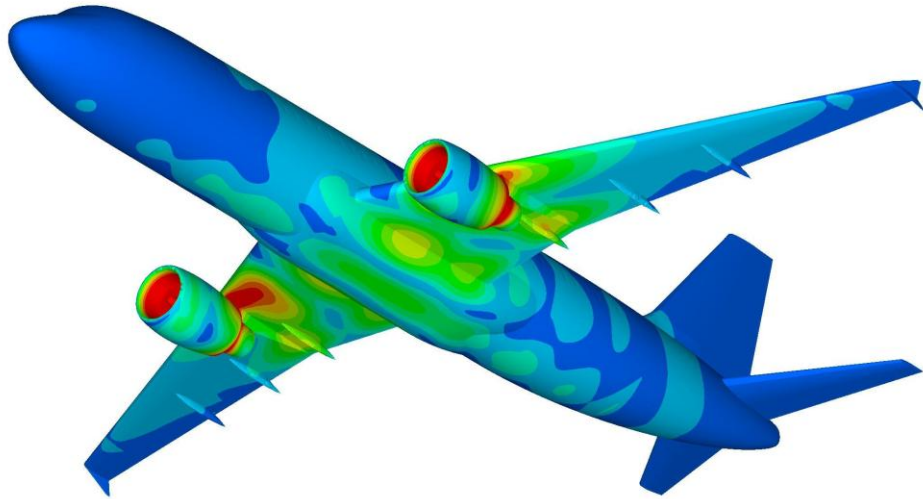
Engineering a control system



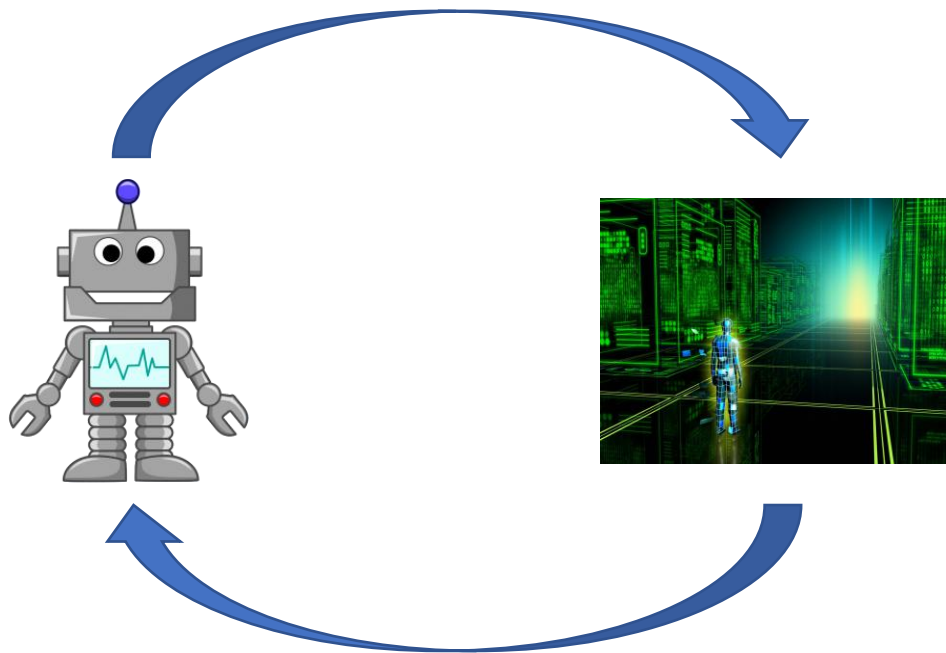
$$\begin{aligned}\mathbf{r} &= \mathbf{r}(t) = r\hat{\mathbf{e}}_r \\ \mathbf{v} &= v\hat{\mathbf{e}}_r + r\frac{d\theta}{dt}\hat{\mathbf{e}}_\theta + r\frac{d\varphi}{dt}\sin\theta\hat{\mathbf{e}}_\varphi \\ \mathbf{a} &= \left(a - r\left(\frac{d\theta}{dt}\right)^2 - r\left(\frac{d\varphi}{dt}\right)^2\sin^2\theta \right)\hat{\mathbf{e}}_r \\ &\quad + \left(r\frac{d^2\theta}{dt^2} + 2v\frac{d\theta}{dt} - r\left(\frac{d\varphi}{dt}\right)^2\sin\theta\cos\theta \right)\hat{\mathbf{e}}_\theta \\ &\quad + \left(r\frac{d^2\varphi}{dt^2}\sin\theta + 2v\frac{d\varphi}{dt}\sin\theta + 2r\frac{d\theta}{dt}\frac{d\varphi}{dt}\cos\theta \right)\hat{\mathbf{e}}_\varphi\end{aligned}$$



Characterization and simulation...



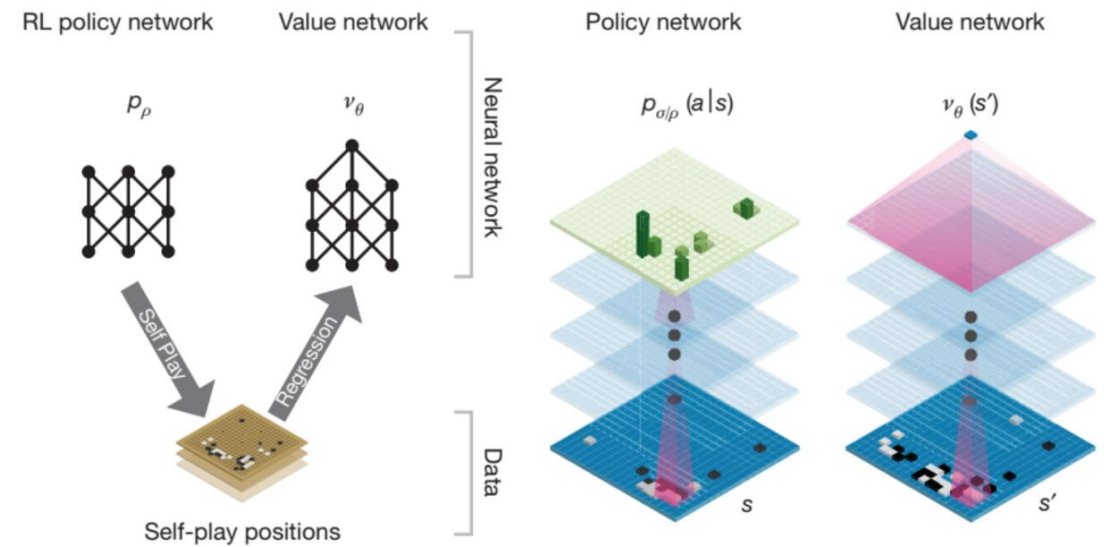
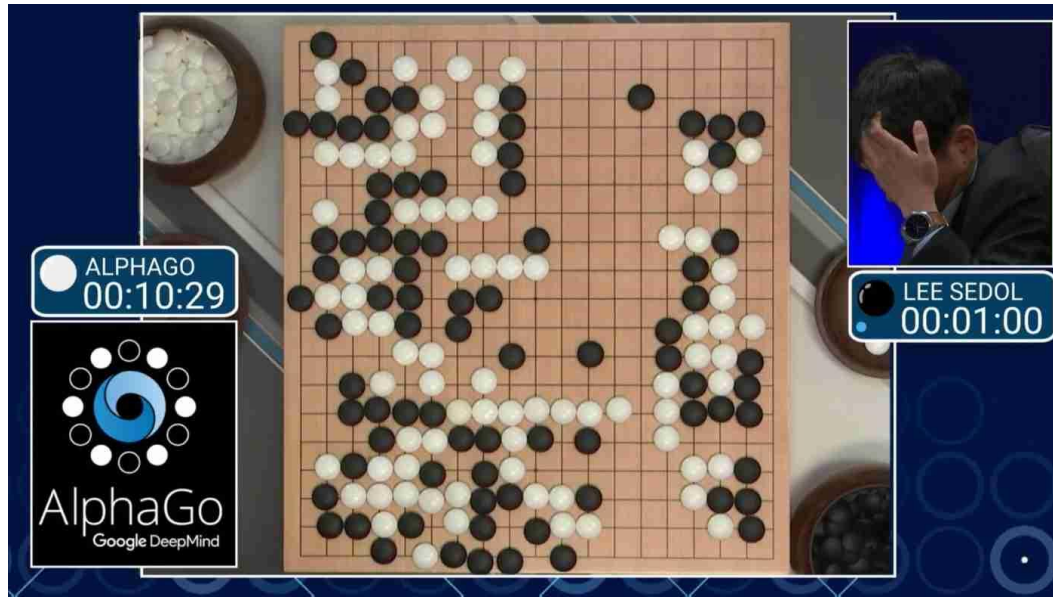
RL: anything you can *simulate* you can *control*



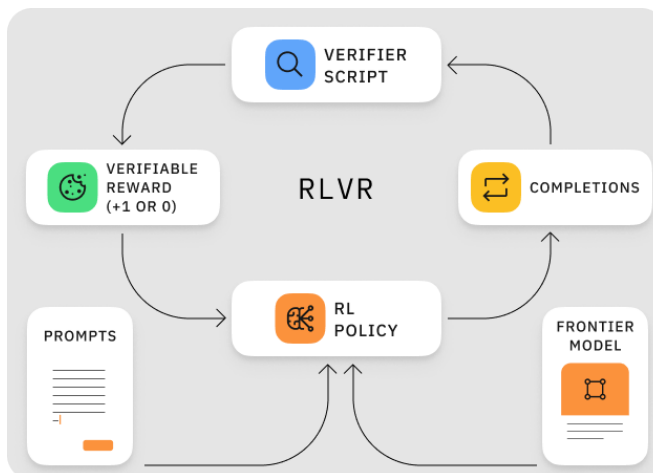
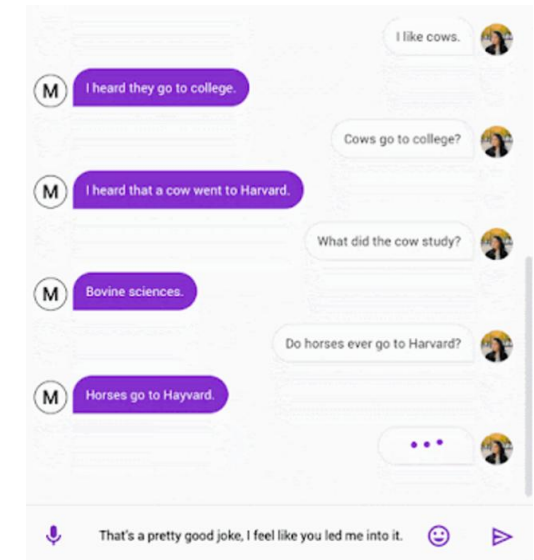
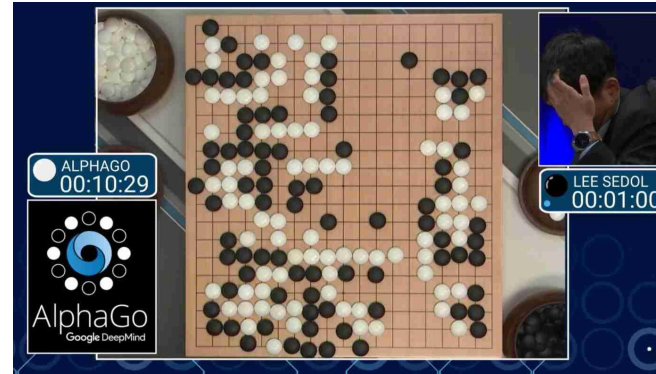
- Provides a powerful engineering tool
- Now *that* different from conventional engineering approach!
 - Before: characterize, simulate, control
 - Now: characterize, simulate, run RL
- Main role: powerful *inversion* engine
- Main weakness: still need to simulate!

Most RL methods in use today in industry are based on this principle!

Example: AlphaGo

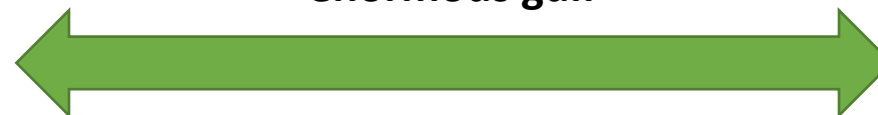


Example: RL for LLMs



even RL for big models (LLMs)
often learns in “simulation”

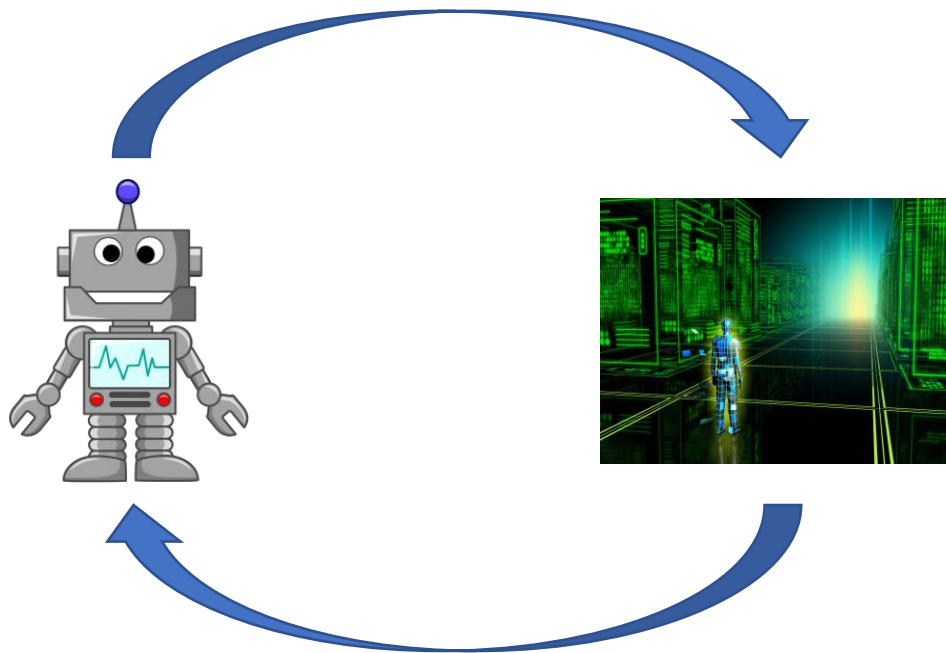
enormous gulf



Example: Sim to real robots



RL: anything you can *simulate* you can *control*



- Provides a powerful engineering tool
- Now *that* different from conventional engineering approach!
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Reinforcement Learning and the Real World

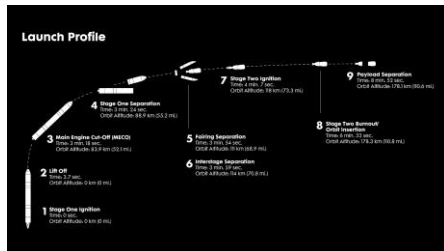


Moravec's paradox

Moravec's paradox seems like a statement about AI

but it is actually a statement about the physical universe

“easy” universes



Why?

“hard” universes



We are all prodigious olympians in perceptual and motor areas, so good that we make the difficult look easy. Abstract thought, though, is a new trick, perhaps less than 100 thousand years old. We have not yet mastered it. It is not all that intrinsically difficult; it just seems so when we do it.

- Hans Moravec

The main lesson of thirty-five years of AI research is that the hard problems are easy and the easy problems are hard. The mental abilities of a four-year-old that we take for granted – recognizing a face, lifting a pencil, walking across a room, answering a question – in fact solve some of the hardest engineering problems ever conceived.

- Steven Pinker



What does this have to do with RL?



How do we engineer a system that can deal with the unexpected?

- Minimal external supervision about what to do
 - Unexpected situations that require adaptation
 - Must discover solutions autonomously
 - Must “stay alive” long enough to discover them!
-
- Humans are extremely good at this
 - Current AI systems are extremely bad at this
 - RL *in principle* can do this, and nothing else can

So what's the problem?



RL *should* be really good in the “hard” universes!

➤ RL *in principle* can do this, and nothing else can

But we rarely study this kind of setting in RL research!

“easy” universes

success = high reward
 (“optimal control”)

closed world, rules
 are known

lots of simulation

Main question: can RL
 algorithms **optimize**
 really well

“hard” universes

success = “survival”
 (“good enough control”)

open world, everything
 must come from data

no simulation (because
 rules are unknown)

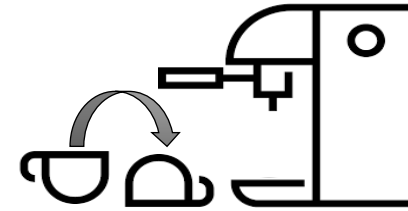
Main question: can RL
 generalize and **adapt**

Some questions that come up in the real world

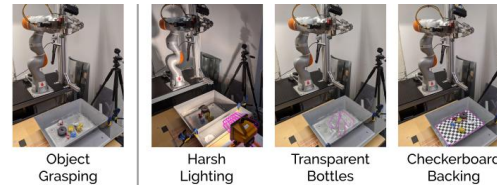
How do we tell RL agents **what we want them to do**?



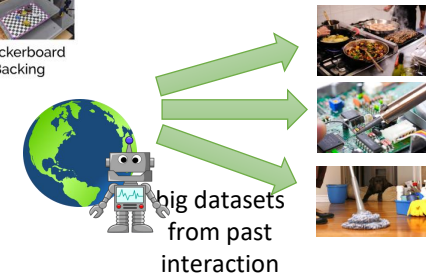
How can we learn **fully autonomously** in continual environments?



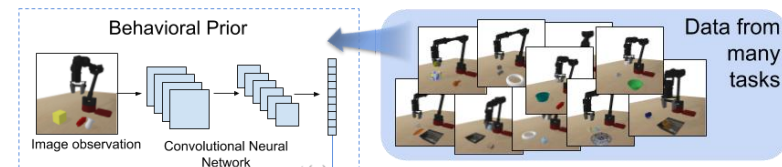
How do remain **robust** as the environment changes around us?



What is the right way to **generalize** using **experience & prior data**?



What is the right way to **bootstrap exploration** with **prior experience**?

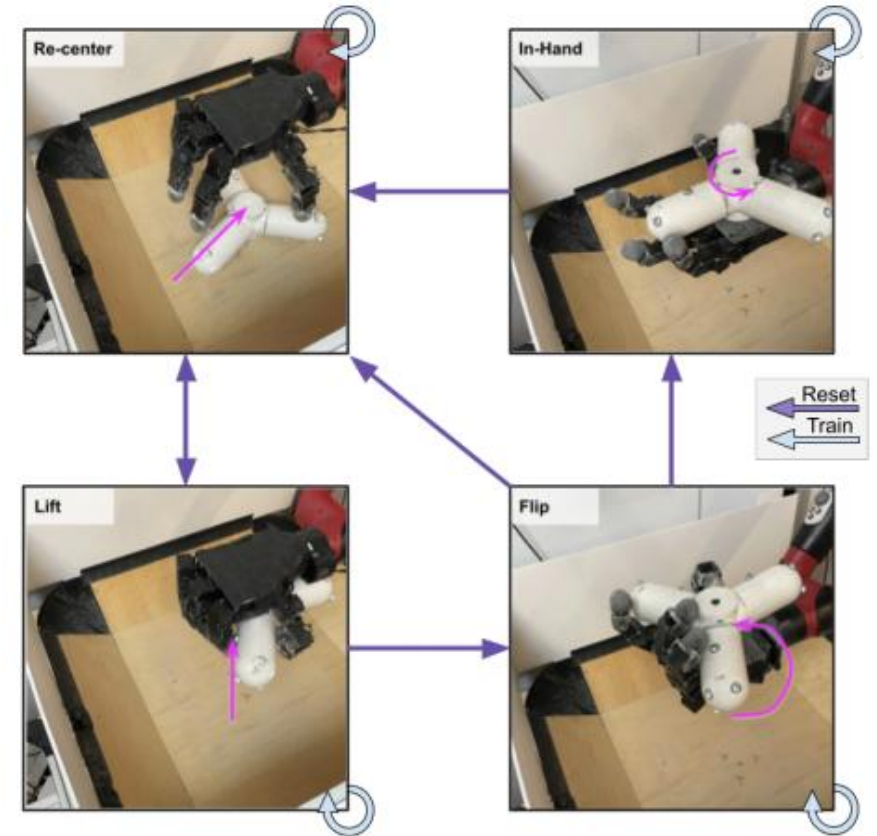


This is not about robots

Robots are the most natural for us to think about, because they are embodied like we are



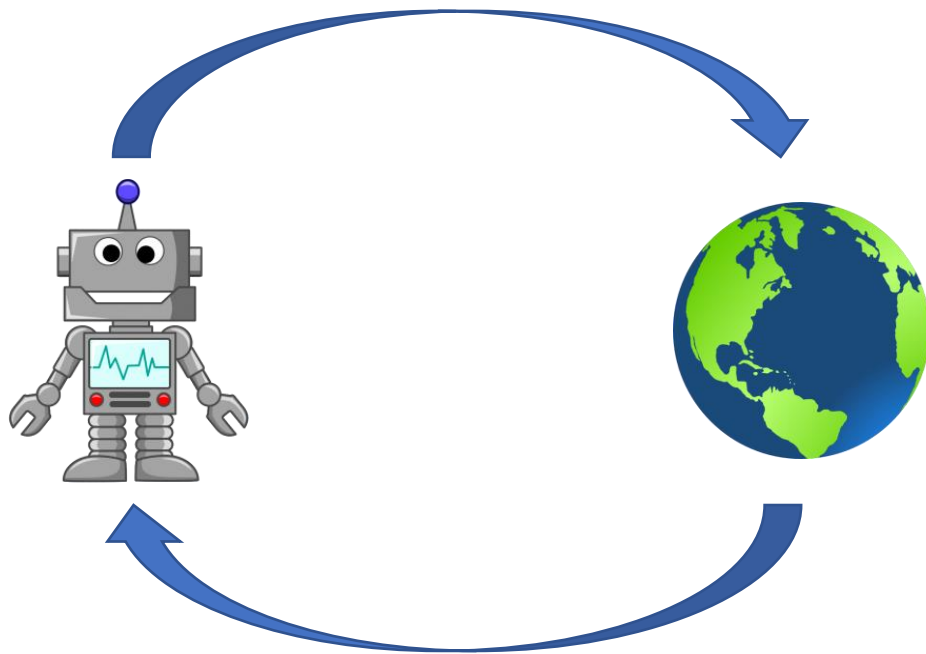
Example: learning autonomously



Example: robot learning with real data



RL in the real world



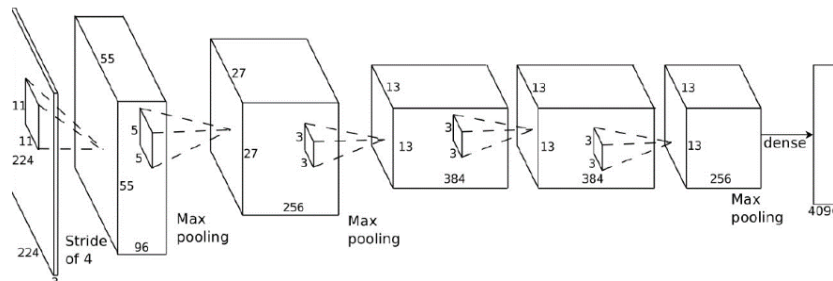
- The real world “simulates itself”: no gap and no train-test mismatch
- Learning under real-world constraints is very hard
- Key to continual learning and self-improving systems
- Where is real-world RL used most often?

This is still the frontier for current RL methods

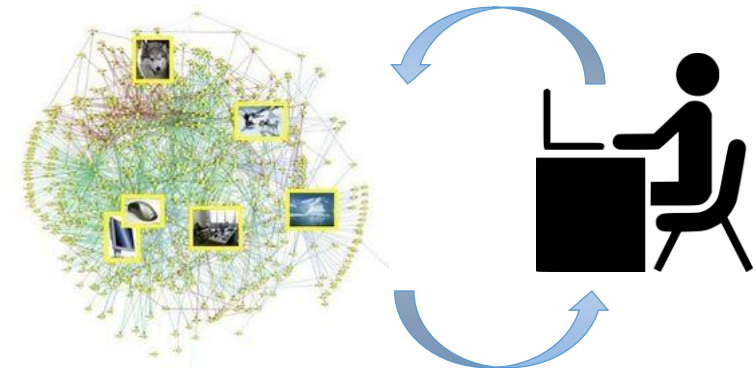
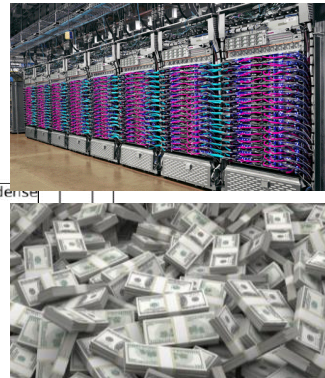
Reinforcement Learning as “Universal” Learning

Large-scale machine learning

Why does deep learning work?

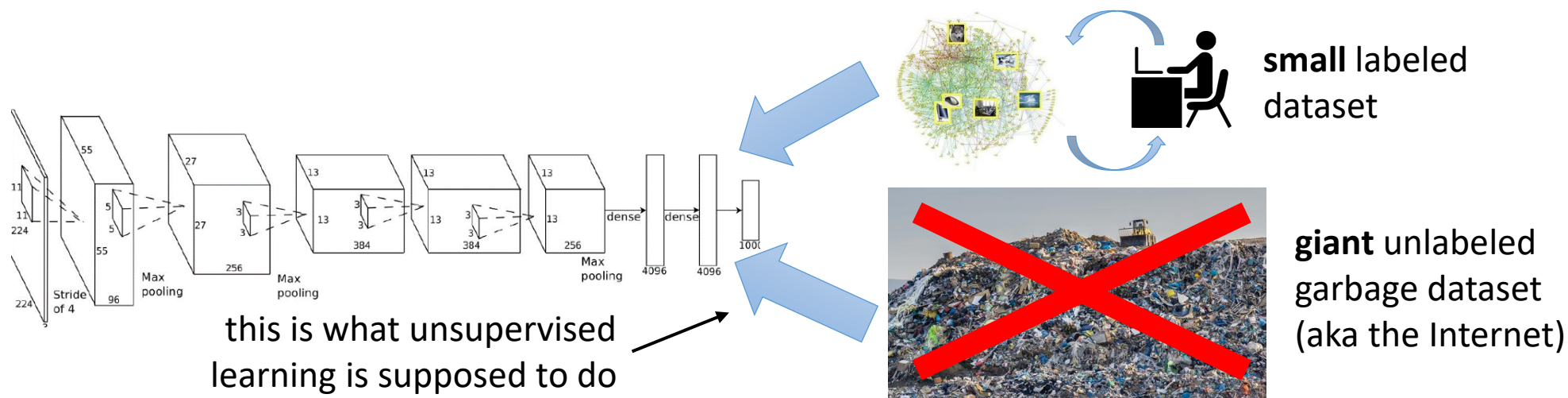


big models



big datasets

Reducing the supervision burden



But then where does the knowledge come from?

“Classic” unsupervised learning: $p_{\theta}(\mathbf{x})$
(this is what, for example, large language models do)



Aside: perhaps this is why “prompting” large language models is such an art!

Stepping back a bit...

Why do we need **machine learning** anyway?

A postulate:

We need machine learning for one reason and one reason only – that's **to produce adaptable and complex decisions.**



Decision: how do I move my joints?



Decision: how do I steer the car?



What is the decision? The image label?

Usually not!

What happens with that label **afterwards**?

Is it used to tag a user's photo?

Detect an endangered animal in a camera trap?

these are **decisions**

they lead to **consequences**

Aside: why do we need **brains** anyway?

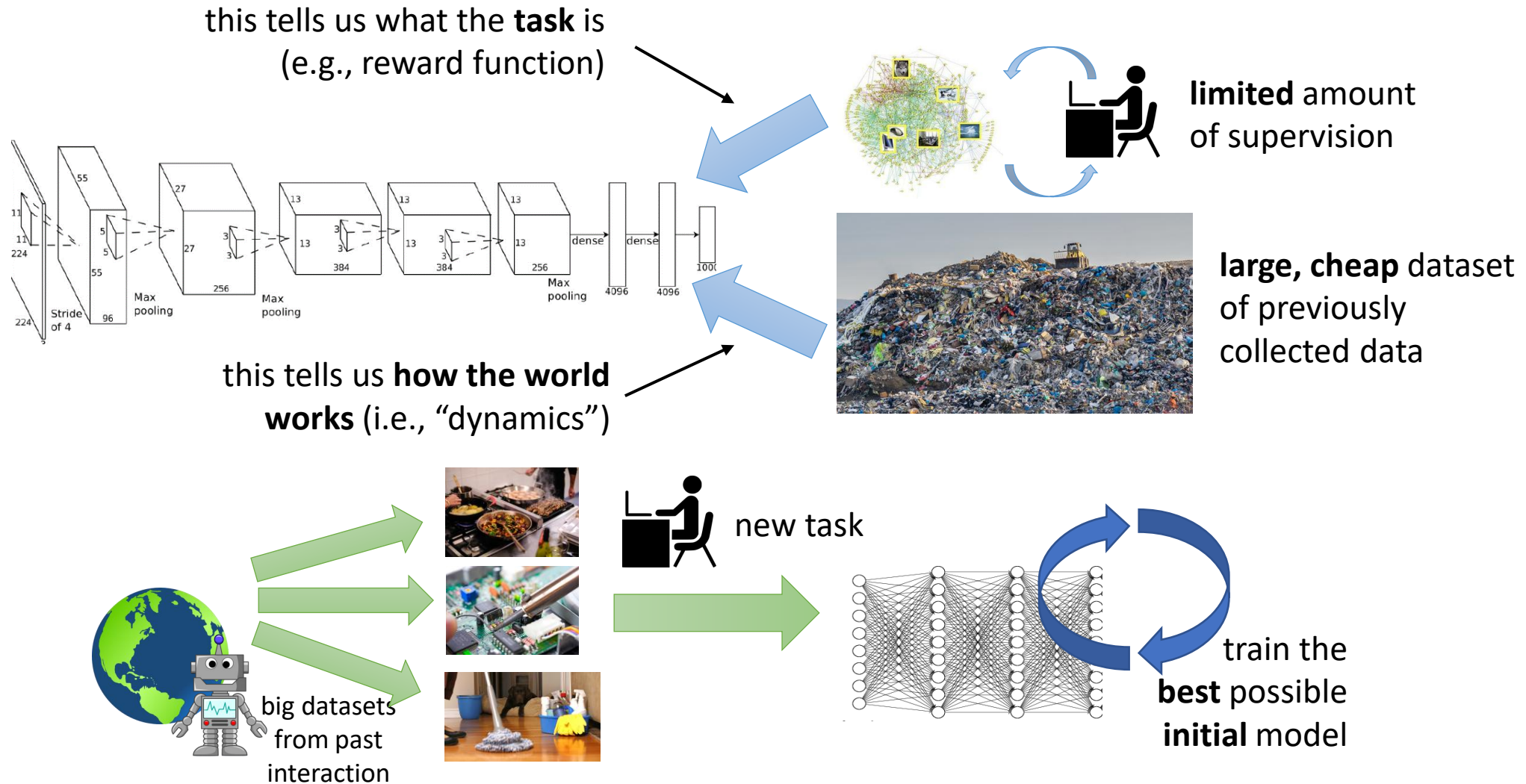


Daniel Wolpert
(knows quite a lot
about brains)

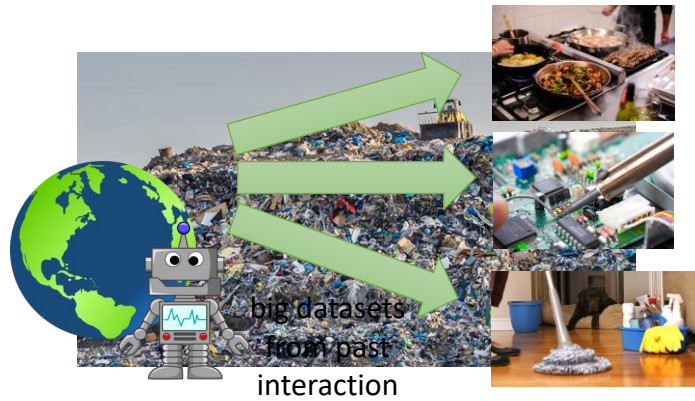


"We have a brain for one reason and one reason only – that's **to produce adaptable and complex movements**. Movement is the only way we have affecting the world around us... I believe that to understand movement is to understand the whole brain."

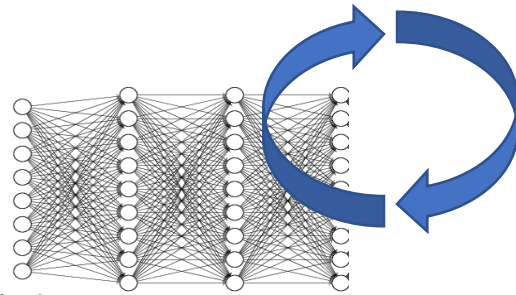
Reinforcement learning as a way to use “cheap” (previously collected) data



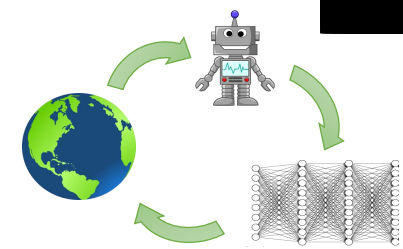
The recipe?



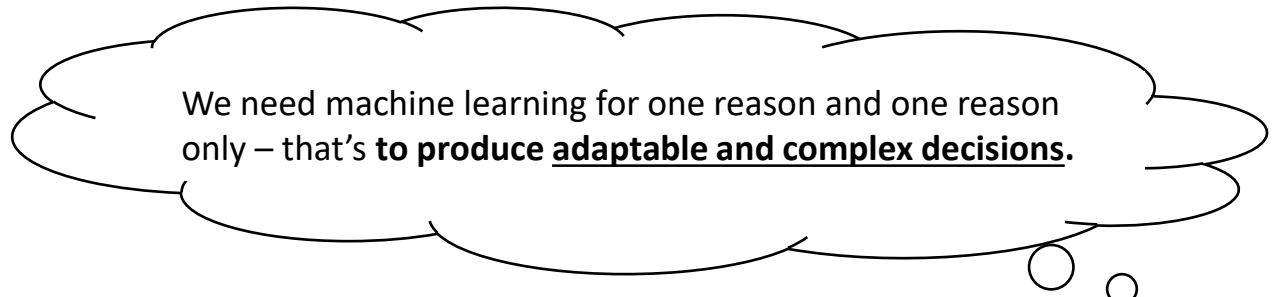
large dataset of diverse
(but possibly low-quality)
behavior



offline?
reinforcement
learning



learning the downstream task



We need machine learning for one reason and one reason only – that's to produce adaptable and complex decisions.

there are a few different
choices here:

- learn a model
- human-defined skills
- goal-conditioned RL
- self-supervised skill discovery

Example: training dialogue agents

What we want:



Can you teach me about behavior cloning?



Of course! I'd be happy to explain behavior cloning. To start, could you tell me if you've ever come across the terms "artificial intelligence" or "machine learning" before?



Yes, I've heard those terms, but I'm not exactly sure what they mean.



No problem at all! Let's take it step by step. Have you ever used a computer or a smartphone?

What we get:



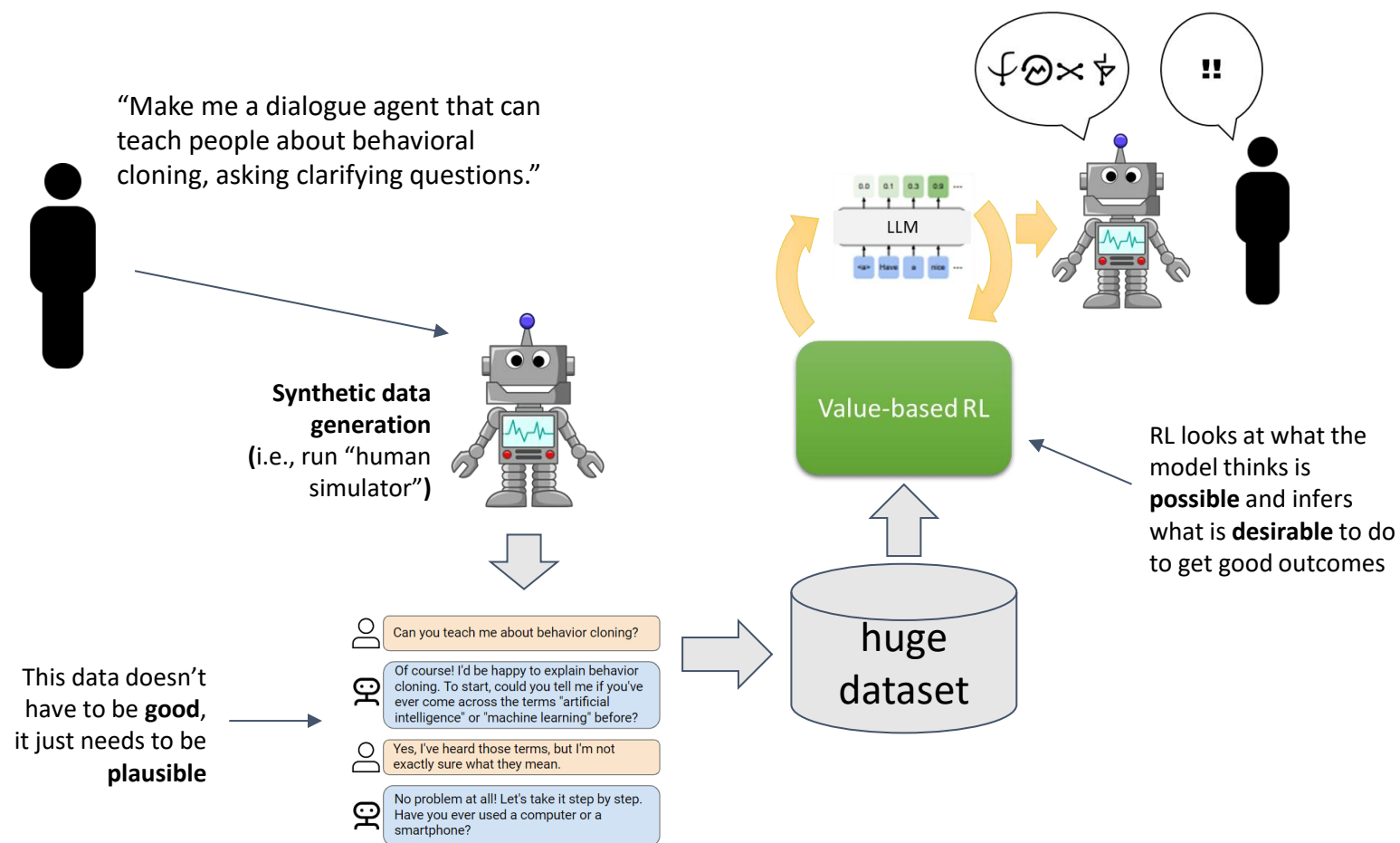
Can you explain the AI concept of behavior cloning to me? But first, I want you to ask me questions to figure out what my background in AI is, then tailor your explanation to match my level.



Certainly! Let's start by understanding your background in AI to provide an explanation that matches your level of knowledge. Please answer the following questions:

1. Have you ever studied or worked with artificial intelligence or machine learning concepts before, even at a basic level?
2. Are you familiar with terms like "neural networks," "training data," or "supervised learning"?
3. Have you ever heard of or worked with specific AI tools or frameworks like TensorFlow, PyTorch, or scikit-learn?
4. Are you interested in a high-level overview or a more detailed explanation of behavior cloning?
5. Do you have any specific examples or applications in mind where you'd like to see behavior

Example: dialogue agents with “model-based” RL



RL + LLM is great for tasks that are easy for LLMs to **simulate** but hard for them to **perform optimally**.

With RL, instead of emulating humans, LLMs can learn to achieve desired outcomes using their understanding of how humans behave.

Example: dialogue agents with “model-based” RL

GPT Agent Dialogue:

H: Hi, I've been hearing about something called "behavior cloning" but I don't really understand what it is. Can you explain it to me?

AI: Certainly! Let's start by understanding your background. Please answer the following questions: Have you ever worked with artificial intelligence or machine learning concepts? Are you familiar with terms like "neural networks," "training data," or "reinforcement learning"? You are interested in a high-level overview or a more detailed explanation?

H: I have some basic knowledge of artificial intelligence and machine learning. I'm an expert. I am familiar with neural networks and supervised learning training data.

AI: Great, thank you for that information. Behavior cloning is a type of machine learning algorithm where a computer learns to perform a task by imitating a human's behavior. It does this by closely analyzing and mimicking the actions of a human expert, using a large dataset of examples. This approach is often used in tasks that don't have a clear set of rules or can't be easily solved through traditional programming methods. Does that make sense so far?

H: Yes, that's starting to make sense. What do you mean by expert?

AI: An expert, in this context, refers to a human or an advanced system that knows how to solve a particular task or problem. The AI model learns the expert's behavior to learn how to perform the task itself. Does that clarify things?

H: Yes, thank you.

IE-RL Agent Dialogue:

H: Hi there! I've heard about this thing called "behavior cloning" in AI, but I don't really understand what it is. Can you explain it to me?

AI: Of course! I'd be happy to explain behavior cloning. To start, could you tell me if you've ever come across the terms "artificial intelligence" or "machine learning" before?

H: Yes, I've heard those terms, but I'm not exactly sure what they mean.

AI: No problem at all! Let's take it step by step. Have you ever used a computer or a smartphone?

H: Yes, I use my computer and phone all the time.

AI: Great! So, imagine that a computer or a smartphone could learn to do something by watching you do it. That's a bit like what behavior cloning is.

H: Oh, interesting. Can you explain more about how that works?

AI: Sure. Behavior cloning is a type of machine learning algorithm where a computer learns to perform a task by imitating a human's behavior. It does this by closely analyzing and mimicking the actions of a human expert, using a large dataset of examples. This approach is often used in tasks that don't have a clear set of rules or can't be easily solved through traditional programming methods. Does that make sense?

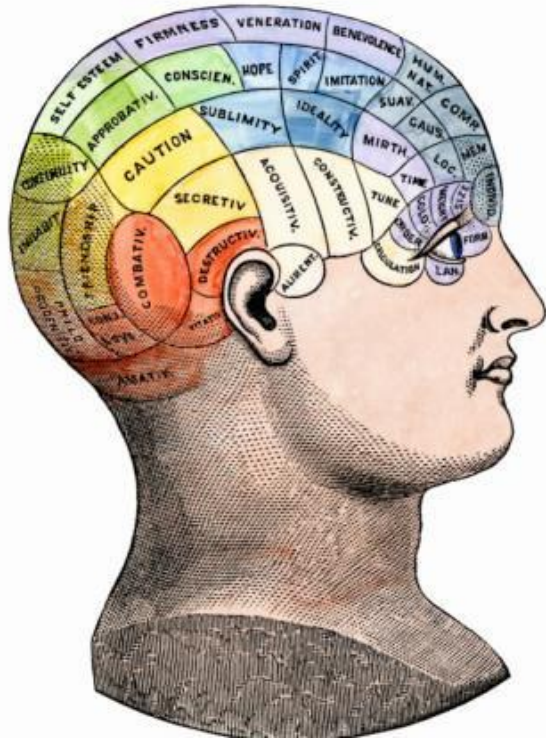
H: Yes, I think so. So how does the computer learn to mimic the behavior?

AI: That's a great question. The computer is fed a large amount of data that shows different examples of the task being performed by a human expert. The computer then uses this data to identify patterns and develop a model of the behavior. This model allows the computer to predict the correct actions to take in a given situation. Over time, the computer can continue to refine its model and improve its performance. Does that help clarify things?

H: That does. Thank you!

Back to the Bigger Picture

Learning as the basis of intelligence



- Reinforcement learning = can reason about decision making
- Deep models = allows RL algorithms to learn and represent complex input-output mappings

Deep models are what allow reinforcement learning algorithms to solve complex problems end to end!

What is missing?

➤ **How Much Information Does the Machine Need to Predict?** Y LeCun

- **"Pure" Reinforcement Learning (cherry)**
 - ▶ The machine predicts a scalar reward given once in a while.
 - ▶ **A few bits for some samples**
- **Supervised Learning (icing)**
 - ▶ The machine predicts a category or a few numbers for each input
 - ▶ Predicting human-supplied data
 - ▶ **10→10,000 bits per sample**
- **Unsupervised/Predictive Learning (cake)**
 - ▶ The machine predicts any part of its input for any observed part.
 - ▶ Predicts future frames in videos
 - ▶ **Millions of bits per sample**



■ (Yes, I know, this picture is slightly offensive to RL folks. But I'll make it up)

Where does the *signal* come from?

- Yann LeCun's cake
 - Unsupervised or self-supervised learning
 - Model learning (predict the future)
 - Generative modeling of the world
 - Lots to do even before you accomplish your goal!
- Imitation & understanding other agents
 - We are social animals, and we have culture – for a reason!
- The giant value backup
 - All it takes is one +1
- All of the above

How should we answer these questions?

- Pick the right problems!
 - Ask: does this have a **chance** of solving an important problem?
 - Optimism in the face of uncertainty is a good exploration strategy!
- Don't be afraid to change the problem statement
 - Many of these challenges won't be met by iterating on existing benchmarks!
- Applications matter
 - Sometimes applying methods to realistic and challenging real-world domains can teach us a lot about the important things that are missing
 - RL has a long history of disregarding this fact
- Think big and start small